MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 12DP7C

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Magnetic deflection and focus, round 12-inch, aluminized, grey spherical faceplate

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Ec1	Ec2	Eb	Ehk	Rg1	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	Meg	ft
Maximum:	6.9	-200	770	13,200	+180	(See note 1)	60,000
Minimum:	5.7	0	---	4,000	-180	---	---
TEST CONDITIONS:	6.3	(See note 2) Adjust	250	4,000	---	---	---

GENERAL:

Qualification - Required

Ⓑ denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Qualification							
1141	Pressure (implosion)		---	---	---	---	---	---
5111	Vibration		---	---	Width	---	2.0	mm
	Quality conformance inspection, part 1		(B)					
5201	Voltage breakdown		0.65	II	---	---	---	---
5201	Voltage breakdown (magnetic types)		0.65	II	---	---	---	---
5206	Gas ratio	See note 3	0.65	II	Gr	---	0.25	---
5101	Neck straightness alinement		0.65	II	---	---	---	---
5106	Screen and faceplate blemishes		0.65	II	---	---	---	---
5223	Modulation	Ib = 200 μA	0.65	II	ΔEc1	---	38	Vdc
5231	Spot position (magnetic deflection)		0.65	II	---	---	20	mm
5236	Zero-bias anode current (magnetic deflection)		0.65	II	---	---	---	---
5241	Grid-cutoff voltage		0.65	II	Ec1	-25	-70	Vdc
5251	Grid No. 1 leakage current		0.65	II	---	---	---	---
---	Aperture alinement	See note 5	0.65	II	---	---	8	mm
	Quality conformance inspection, part 2							
1101	Secureness of base, cap, or insert		---	---	---	---	---	---
1301	Heater current		---	---	If	540	660	mA
5201	Electrode current	Ec1 = 0	---	---	Ic2	---	50	μAdc
5101	Side terminal and base alinement	Pin No. 5	---	---	---	---	---	---
5101	Neck and bulb alinement (magnetic types)		---	---	---	---	---	---
5101	Face tilt		---	---	---	---	---	---
5216	Stray light emission (conventional types)	Eb2 = 13,200 Vdc; Ec2 = 770 Vdc	---	---	---	---	---	---
5221	Screen		---	---	---	---	---	---
5221	Screen brightness I	See note 4	---	---	---	---	---	---
5226	Line width A (magnetic deflection)	Ib = 200 μAdc	---	---	Width	---	0.50	mm

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued							
5226	Line width C (magnetic deflection)	ib = 200 μAdc	---	---	Width	---	0.75	mm
5246	Focusing ampere turns	Ib = 200 μAdc; D = 4.125 inch	---	---	---	337	459	---
5251	Heater-cathode leakage current		---	---	---	---	---	---
5251	Grid No. 2 leakage current		---	---	---	---	---	---
(B)1111	Base pin solder depth		---	---	---	---	---	---
	Quality conformance inspection, part 3							
---	Life-test provisions	Group C; t = 500 hours (min); Eb2 = 13, 200 Vdc; Ec2 = 770 Vdc; Ib = 60 μAdc	---	---	---	---	---	---
---	Life-test end points	Line width A; Ib = 200 μAdc	---	---	Width	---	0.50	mm
		Line Width C; Ib = 200 μAdc	---	---	Width	---	0.75	mm
		Modulation; Ib = 200 μAdc	---	---	ΔEc1	---	38	Vdc
(B)	Periodic-check tests							
1331	Direct-interelectrode capacitance	k to all	---	---	Ck	---	10	pF
		g1 to all	---	---	Cg1	---	11	pF
		g2 to all	---	---	Cg2	---	12	pF
5216	Cathode illumination		---	---	---	---	---	---

NOTES:

1. When Ec2 is greater than 330 Vdc, Rg1 shall not exceed 0.5 Meg. When Ec2 is less than 330 Vdc, Rg1 shall not exceed 1.5 Meg.
2. When Ec2 is greater than 330 Vdc or Ec1 is near zero, the effective resistance of the anode and grid No. 3 supply should be adequate to limit the anode and grid No. 3 input power to 6 watts. The peak grid No. 1 drive from cutoff should never exceed 65 volts.
3. This test to be performed at the conclusion of the holding period.
4. Screen brightness I shall be measured as specified in method 5221 except that no pulsing shall be used on the grid and the screen current shall be 26.7 μA with an anode voltage of 9kV. The steady-state output shall be 735 cb minimum.
5. The distance between the center of the unfocused, undeflected spot at low intensity (Ec1 near cut-off) and the center of the image of the masking aperture observed at high intensity of the unfocused, undeflected spot shall not exceed the limit specified. Ec1 should not be held at zero for more than approximately 30 seconds to prevent damage to the screen.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL,
Navy -
Air Force - 11, 17, 85
DSA - ES

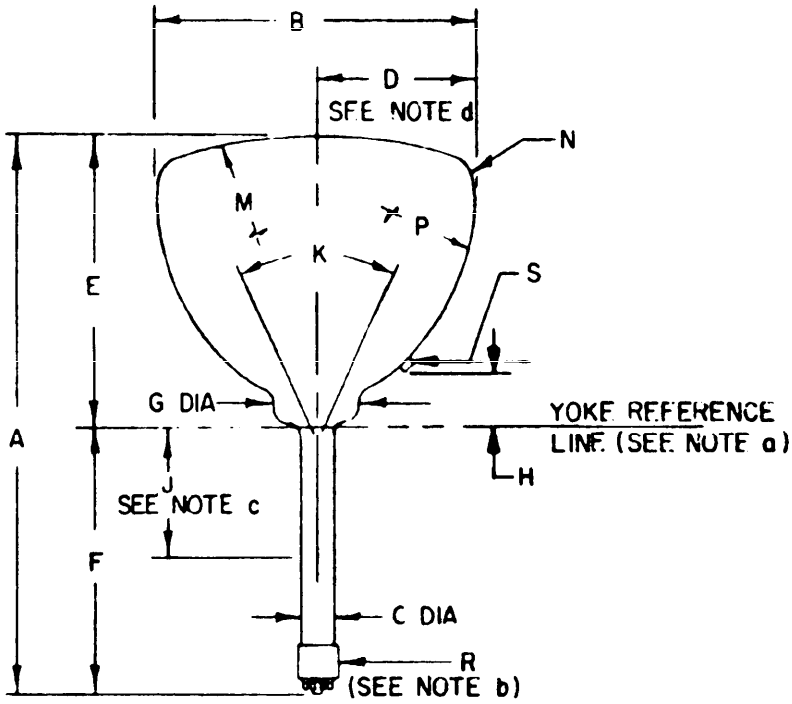
User activities:
Army - WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

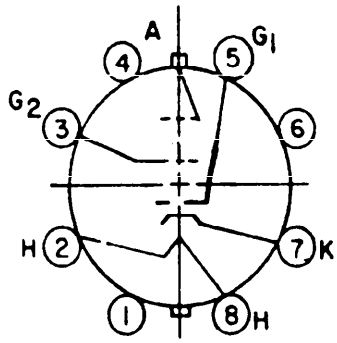
Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5960-2401-54)

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BASING DIAGRAM

NOTES:
a. Reference line is determined by the plane of the upper edge of the reference line gage (JEDEC No. 112) when the gage is resting on the cone.
b. Anode terminal aligns with pin No. 5 ± 10 degrees.
c. Recommended position for center of focusing field.
d. Useful screen radius.
e. Reference dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube type 12DP7C.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	19.250 (488.9)	20.000 (508.00)
B	11.813 (300.05)	12.187 (309.55)
C	1.313 DIA (33.35)	1.437 DIA (36.50)
D	5.000 (127.00)	
E	10.313 (261.95)	10.687 (271.45)
F	8.937 (227.00)	9.312 (236.52)
G	3.063 DIA (77.80)	3.187 DIA (80.98)
H	1.625 (41.25)	2.125 (53.98)
Quality conformance inspection, part 3 (periodic check)		
R	BASE: B8-65 (EIA)	
S	Bulb contact: C1-5	
Reference dimensions (see note e)		
J	4.125 (104.78)	
K	50°	
M	20.000 R (508.00)	
N	1.000 R (25.40)	
P	7.687 R (195.25)	

FIGURE 1. Outline drawing of electron tube type 12DP7C - Continued.

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Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-E-1/1222B, Electron Tube, Cathode Ray Type 12DP7C

ORGANIZATION

CITY AND STATE	CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*	DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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